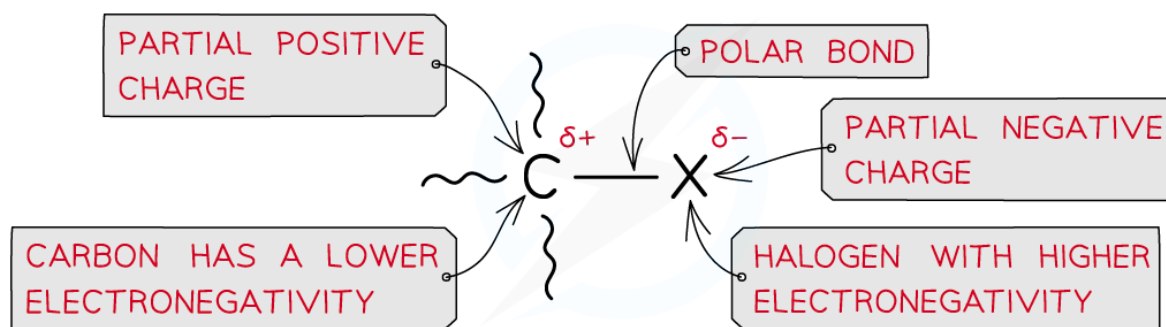


Substitution Reactions of Halogenoalkanes (Cambridge (CIE) A Level Chemistry): Revision Note

Nucleophilic Substitution Reactions of Halogenoalkanes

- **Halogenoalkanes** are much more reactive than alkanes due to the presence of the **electronegative** halogens
 - The halogen-carbon bond is polar causing the carbon to carry a partial positive and the halogen a partial negative charge
- A **nucleophilic substitution** reaction is one in which a **nucleophile** attacks a carbon atom which carries a **partial positive charge**
- An atom that has a **partial negative charge** is replaced by the nucleophile

Explaining the polarity of a carbon-halogen bond

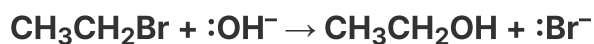


Due to large differences in electronegativity between the carbon and halogen atom, the C-X bond is polar

Reaction with NaOH

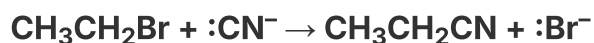
- The reaction of a halogenoalkane with **aqueous alkali** results in the formation of an alcohol
- The halogen is **replaced** by the OH^-
- The **aqueous hydroxide** (OH^- ion) behaves as a **nucleophile** by donating a pair of electrons to the carbon atom bonded to the halogen

- For example, bromoethane reacts with aqueous alkali when heated to form ethanol
 - Hence, this reaction is a **nucleophilic substitution**
 - The halogen is replaced by a nucleophile, :OH^-



Reaction with KCN

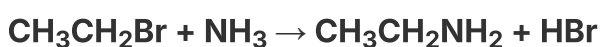
- The **nucleophile** in this reaction is the **cyanide**, CN^- ion
- Ethanolic solution** of **potassium cyanide** (KCN in ethanol) is **heated under reflux** with the halogenoalkane
- The product is a **nitrile**
 - For example, bromoethane reacts with ethanolic potassium cyanide when heated under reflux to form propanenitrile
 - The halogen is replaced by a nucleophile, :CN^-



- The nucleophilic substitution of halogenoalkanes with KCN adds an **extra** carbon atom to the carbon chain
- This reaction can therefore be used by chemists to make a compound with one more carbon atom than the best available organic starting material

Reaction with NH_3

- The **nucleophile** in this reaction is the **ammonia**, NH_3 molecule
- An **ethanolic solution** of **excess ammonia** (NH_3 in ethanol) is **heated under pressure** with the halogenoalkane
- For example, bromoethane reacts with excess ethanolic ammonia when heated under pressure to form ethylamine
 - The product is a **primary amine**
 - The halogen is replaced by an amine group NH_2

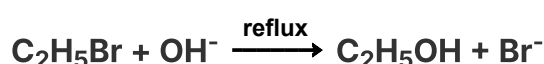


- It is very important that the ammonia is in excess as the product of the nucleophilic substitution reaction, the **ethylamine**, can act as a nucleophile and attack another bromoethane to form the **secondary amine, diethylamine**

Reaction with aqueous silver nitrate

- Halogenoalkanes can be broken down **under reflux** by water to form **alcohols**
- The breakdown of a substance by water is also called **hydrolysis**

- The water in aqueous silver nitrate will hydrolyse the halogenoalkane
 - The fastest nucleophilic substitution reactions take place with the iodoalkanes as the C-I bond is the **weakest (longest)**
 - The slowest nucleophilic substitution reactions take place with the fluoroalkanes as the bond is the **strongest (shortest)**
- For example, bromoethane reacts with aqueous silver nitrate solution to form ethanol and a Br⁻ ion
 - The Br⁻ ion will form a cream precipitate with Ag⁺
- This reaction is classified as a nucleophilic substitution reaction with **water molecules** in aqueous **silver nitrate** solution acting as nucleophiles, **replacing** the halogen in the halogenoalkane



Nucleophilic substitution with OH⁻



In nucleophilic substitution with OH⁻, the bond that forms and the bond that breaks must both involve the carbon atom that is bonded to the leaving group

- This reaction is similar to the nucleophilic substitution reaction of halogenoalkanes with aqueous alkali, however, hydrolysis with water is much **slower** than with the OH⁻ ion in alkalis
 - The hydroxide ion is a better nucleophile than water as it carries a full formal negative charge
 - In water, the oxygen atom only carries a partial negative charge

Comparing water and the hydroxide ion as nucleophiles



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A hydroxide ion is a better nucleophile as it has a full formal negative charge whereas the oxygen atom in water only carries a partial negative charge; this causes the nucleophilic substitution reaction with water to be much slower than with aqueous alkali

- The halogenoalkanes have different **rates of hydrolysis**
- This reaction can be used as a **test** to identify halogens in a halogenoalkane by measuring how long it takes for the test tubes containing the halogenoalkane and aqueous silver nitrate solutions to become **opaque**